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SUPPLEMENTARY MATERIAL

Supplementary numerical audit

How informative is agreement among nuclear-mass models? Chronology-respecting tests of error coverage and shell proximity

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S1. Scope of independent reproduction

The companion `audit_analysis.py` reads the unchanged original standardized mass-excess and historical eligibility tables bundled in the accompanying `v1.2.1` scientific package, including the GSM revision addendum. It forms the cohorts from their original membership flags, joins four prediction tables by (N, Z) and independently computes the analysis. It reproduces all 51 archived primary-confirmation mass-error and 34 separation-energy-error scores to numerical precision. It reconstructs the predeclared broad population without applying the 100 keV experimental-uncertainty cutoff. The original conversion of source AME and theoretical model files was not scripted in the original archive. New independent source parsers reproduce all 10,347 AME2012/2016/2020 standardized entries and the complete archived FRDM2012 and WS4 rows from official inputs; HFB-27 matches the independent RIPL-4 copy within 5 keV. The original AMDC `dz31.f` source was subsequently compiled and executed on an author's Windows system using GNU Fortran 16.0.1 with `-std=legacy -ffixed-line-length-none -O0`. The accompanying independent Fortran checker compared every one of the 21,275 printed DZ31 mass-excess values with the unchanged archived CSV by (N, Z) and A. The supplied original execution report records a maximum absolute difference of 0.000 MeV at the archive's 0.01-MeV output precision; all differences were within the stated 0.011-MeV acceptance tolerance. The report and checking code are preserved in `raw_source_audit/`, along with their SHA-256 provenance. In the further v9 archive audit, the original unmodified 957,375-byte `massex.out` was recovered from an earlier revision archive. A second, independent fixed-format parser checks all 21,275 lines against the unchanged DZ31 CSV: every (N, Z) key and $A = N + Z$ match and the maximum absolute difference is 0.000 MeV at printed precision. The raw file SHA-256 is `32ad383091ef86d4ed5af28b2e4ad4caab5cee6bb2194ebae1cb83d0431ffb61`. This validates source-to-table agreement *at printed precision*, not unrounded floating-point identity or the unknown historical fit roster. The original BRUSLIB-to-CSV HFB-27 transformation remains unavailable.

The per-target quantities are $D = \mathrm{sd}(M_1, \ldots, M_4)$ (sample standard deviation, with denominator 3) and $E = |\frac{1}{4}\sum_k M_k - M_{\mathrm{exp}}|$. Coverage of $\pm cD$ means simply $E \leq cD$. Reported uncertainty intervals are descriptive. Percentile bootstrap intervals use 10,000 deterministic draws, with seed 42 for rank correlations, within-bin summaries, MAE, coverage and shell contrasts, and seed 53 for the RMS scale ratio. The isotopic-chain bootstrap draws entire unique-Z blocks with replacement, preserving the original block sizes within a draw. Chain-block intervals are available for MAE, $C(D)$, $C(2D)$, ρ and the RMS ratio. Some intervals are therefore broad.

S2. Primary and prespecified sensitivity cohorts

Table 1: Mass-excess results; rho confidence intervals use Z-chain block resampling.

Cohort	n	ρ	p	block 95% CI	MAE	k_{RMS}	$E \leq D$	$E \leq 2D$
Development, primary	32	0.422	0.0162	[-0.005,0.740]	0.403	1.255	25/32	31/32
Development, broad	58	0.402	0.0018	[0.042,0.693]	0.449	1.046	42/58	53/58
Confirmation, primary	51	0.265	0.0604	[-0.057,0.558]	0.247	0.703	42/51	49/51
Confirmation, broad	71	0.329	0.0051	[0.055,0.566]	0.303	0.818	57/71	68/71

Table 2: Wilson intervals are binomial descriptive bounds; scale-ratio intervals are Z-chain bootstraps. Continuous ranked probability score (CRPS) values are in MeV; lower values indicate better predictive performance.

Cohort	$C(D)$	95% CI	$C(2D)$	95% CI	k block CI	CRPS(D)	CRPS(const.)
Development, primary	0.781	[0.612,0.890]	0.969	[0.843,0.994]	[0.648,1.478]	0.290	0.361
Development, broad	0.724	[0.598,0.822]	0.914	[0.814,0.963]	[0.699,1.289]	0.325	0.353
Confirmation, primary	0.824	[0.697,0.904]	0.961	[0.868,0.989]	[0.537,0.882]	0.198	0.227

Cohort	$C(D)$	95% CI	$C(2D)$	95% CI	k block CI	CRPS(D)	CRPS(const.)
Confirmation, broad	0.803	[0.696,0.879]	0.958	[0.883,0.986]	[0.593,1.024]	0.234	0.258

The Wilson intervals in Table 2 treat nuclei as independent binomial observations and are retained as descriptive single-observation bounds. The intervals in Table 3 instead resample *entire isotopic chains* (unique Z, 10,000 draws, seed 42), providing the clustering-aware uncertainty requested by the referee. These percentile intervals preserve within-chain sample sizes but do not resolve concentration in a few nuclear-chart regions.

Table 3: 95% Z-chain percentile-bootstrap intervals for three headline outcomes; MAE in MeV, coverages as proportions.

Cohort	Z chains	MAE block CI	$C(D)$ block CI	$C(2D)$ block CI
Development, primary	23	[0.189,0.736]	[0.607,0.929]	[0.885,1.000]
Development, broad	39	[0.298,0.645]	[0.576,0.855]	[0.800,1.000]
Confirmation, primary	30	[0.192,0.323]	[0.690,0.923]	[0.894,1.000]
Confirmation, broad	37	[0.227,0.403]	[0.697,0.890]	[0.909,1.000]

S3. Frozen development-quintile diagnostics

The six original quintile boundaries were fixed in primary development: D in MeV, approximately 0.127772, 0.230000, 0.328934, 0.424490, 0.514996 and 1.600383. Operationally, the interior four edges are reused; exterior bounds are open, so out-of-range broad cases remain assigned. CI bounds are *ordinary* per-nucleus bootstrap intervals, not cluster-robust intervals, because certain bins include few chains.

Table 4: Within-bin mean absolute error intervals and RMS error/spread (MeV).

Cohort	Bin	n	mean E	68% CI	95% CI	RMS(E)	RMS(D)
Development, primary	Q1	7	0.249	[0.103,0.394]	[0.062,0.549]	0.436	0.191
Development, primary	Q2	6	0.183	[0.135,0.232]	[0.091,0.279]	0.219	0.272
Development, primary	Q3	6	0.160	[0.123,0.197]	[0.087,0.231]	0.184	0.391
Development, primary	Q4	6	0.241	[0.176,0.307]	[0.117,0.369]	0.291	0.453
Development, primary	Q5	7	1.090	[0.722,1.458]	[0.427,1.831]	1.449	1.067
Development, broad	Q1	9	0.343	[0.220,0.464]	[0.131,0.600]	0.498	0.191
Development, broad	Q2	7	0.203	[0.157,0.247]	[0.115,0.293]	0.235	0.275
Development, broad	Q3	8	0.292	[0.195,0.390]	[0.133,0.500]	0.398	0.388
Development, broad	Q4	11	0.329	[0.239,0.418]	[0.182,0.522]	0.438	0.468
Development, broad	Q5	23	0.676	[0.538,0.815]	[0.433,0.969]	0.939	0.922
Confirmation, primary	Q1	6	0.245	[0.169,0.321]	[0.107,0.411]	0.309	0.192
Confirmation, primary	Q2	7	0.116	[0.079,0.151]	[0.053,0.192]	0.150	0.296
Confirmation, primary	Q3	17	0.185	[0.147,0.224]	[0.117,0.265]	0.243	0.379
Confirmation, primary	Q4	10	0.275	[0.223,0.327]	[0.179,0.381]	0.320	0.465
Confirmation, primary	Q5	11	0.402	[0.309,0.496]	[0.223,0.592]	0.507	0.731
Confirmation, broad	Q1	7	0.258	[0.189,0.326]	[0.136,0.394]	0.313	0.180
Confirmation, broad	Q2	10	0.161	[0.127,0.195]	[0.096,0.226]	0.193	0.299
Confirmation, broad	Q3	20	0.173	[0.138,0.206]	[0.112,0.244]	0.229	0.381
Confirmation, broad	Q4	10	0.275	[0.223,0.327]	[0.179,0.381]	0.320	0.465
Confirmation, broad	Q5	24	0.495	[0.407,0.584]	[0.339,0.681]	0.650	0.754

S4. Complete 71-nucleus AME2016–AME2020 confirmation roster

The primary cohort comprises the 51 rows marked "Y"; the other 20 are the additional broad cases. D, E and experimental mass-excess uncertainty are expressed in MeV and keV as indicated. Distance d_1 is Manhattan and d_2 Euclidean, measured to a primary-eligible AME2016 nucleus. The near-shell flag is one if N or Z lies within two of a specified major shell closure. Further precision, including the S_{2n} columns, is preserved in the companion CSV.

Table 5: All scored confirmation nuclei; full-precision values are provided in the companion CSV.

Nucleus	N	Z	Primary?	σ_{mexp} (keV)	D (MeV)	E (MeV)	Shell	d_1	d_2
30Cl	13	17	Y	23.9	0.195	0.393	0	1	1.000
55Ca	35	20	N	160.2	1.013	1.915	1	1	1.000
56Ca	36	20	N	249.6	0.896	0.765	1	2	2.000
58Sc	37	21	N	190.0	0.749	0.414	1	4	2.828
59Sc	38	21	N	249.6	1.162	0.114	1	4	3.606
58Ti	36	22	N	183.3	0.579	0.931	1	2	1.414
60Ti	38	22	N	240.3	0.926	0.969	1	3	3.000
62V	39	23	N	264.3	0.298	0.260	0	2	2.000
63V	40	23	N	340.0	0.587	0.420	0	2	2.000
44Cr	20	24	Y	51.2	0.786	0.833	1	1	1.000
46Mn	21	25	Y	86.6	0.728	0.557	1	1	1.000
48Fe	22	26	Y	92.2	0.814	0.528	1	1	1.000
50Co	23	27	N	125.8	0.573	0.057	1	1	1.000

Nucleus	N	Z	Primary?	σ_{mexp} (keV)	D (MeV)	E (MeV)	Shell	d_1	d_2
70Co	43	27	Y	11.0	0.362	0.474	1	1	1.000
52Ni	24	28	Y	82.9	0.471	0.600	1	1	1.000
77Cu	48	29	Y	1.2	0.210	0.588	1	1	1.000
79Cu	50	29	N	105.0	0.593	0.585	1	1	1.000
84Ga	53	31	Y	29.8	0.869	0.029	0	1	1.000
85Ga	54	31	Y	37.3	0.827	0.218	0	2	1.414
73Rb	36	37	Y	40.8	0.202	0.020	0	1	1.000
101Rb	64	37	Y	20.5	0.482	0.509	0	1	1.000
102Rb	65	37	Y	82.9	0.666	0.361	0	2	1.414
89Ru	45	44	Y	24.2	0.426	0.343	0	1	1.000
92Pd	46	46	N	345.0	0.403	0.033	0	2	1.414
93Pd	47	46	N	370.0	0.326	0.294	0	1	1.000
97Cd	49	48	N	420.2	0.779	0.159	1	1	1.000
132Cd	84	48	Y	60.1	0.556	0.988	1	2	1.414
101In	52	49	Y	11.7	0.609	0.592	1	1	1.000
104Te	52	52	N	317.6	0.291	0.250	1	3	2.236
114I	61	53	Y	20.0	0.101	0.133	0	1	1.000
108Xe	54	54	N	379.5	0.613	0.637	0	3	2.236
150Ba	94	56	Y	5.7	0.344	0.594	0	2	1.414
156Pr	97	59	Y	1.0	0.373	0.021	0	1	1.000
157Pr	98	59	Y	3.2	0.210	0.157	0	2	1.414
158Nd	98	60	Y	1.3	0.318	0.212	0	1	1.000
159Nd	99	60	Y	29.8	0.265	0.072	0	2	1.414
160Nd	100	60	Y	46.6	0.313	0.298	0	3	2.236
160Pm	99	61	Y	2.0	0.396	0.225	0	1	1.000
161Pm	100	61	Y	9.0	0.335	0.043	0	2	1.414
162Sm	100	62	Y	3.5	0.388	0.295	0	1	1.000
163Sm	101	62	Y	7.4	0.437	0.058	0	2	1.414
164Sm	102	62	Y	4.1	0.438	0.259	0	3	2.236
164Eu	101	63	Y	2.1	0.472	0.297	0	1	1.000
165Eu	102	63	Y	5.2	0.413	0.150	0	2	2.000
164Gd	100	64	Y	1.0	0.376	0.142	0	1	1.000
165Gd	101	64	Y	1.3	0.498	0.142	0	1	1.000
166Gd	102	64	Y	1.6	0.420	0.229	0	2	1.414
167Gd	103	64	Y	5.2	0.398	0.014	0	3	2.236
165Tb	100	65	Y	1.5	0.279	0.113	0	1	1.000
167Tb	102	65	Y	1.9	0.323	0.031	0	1	1.000
168Tb	103	65	Y	4.2	0.412	0.236	0	2	2.000
211Pa	120	91	Y	69.5	0.476	0.229	0	1	1.000
220Pa	129	91	Y	14.7	0.366	0.104	0	1	1.000
222Pa	131	91	Y	86.6	0.390	0.062	0	1	1.000
220Np	127	93	Y	30.7	0.209	0.178	1	1	1.000
222Np	129	93	Y	38.1	0.563	0.057	0	1	1.000
223Np	130	93	Y	82.9	0.393	0.061	0	1	1.000
224Np	131	93	Y	28.9	0.337	0.320	0	1	1.000
226Np	133	93	N	102.1	0.079	0.335	0	1	1.000
248Bk	151	97	Y	50.1	0.284	0.023	0	1	1.000
246Es	147	99	Y	89.9	0.287	0.061	0	1	1.000
249Md	148	101	N	164.4	0.350	0.152	0	1	1.000
250Md	149	101	Y	90.9	0.348	0.070	0	1	1.000
252Md	151	101	Y	91.3	0.373	0.107	0	1	1.000
253Lr	150	103	N	164.5	0.424	0.120	0	1	1.000
254Lr	151	103	Y	91.3	0.469	0.094	0	1	1.000
257Db	152	105	N	164.6	0.530	0.265	0	1	1.000
258Db	153	105	Y	91.9	0.476	0.218	0	1	1.000
261Bh	154	107	N	179.8	0.748	0.240	0	1	1.000
262Bh	155	107	Y	93.1	0.660	0.124	0	1	1.000
266Mt	157	109	Y	96.5	0.867	0.134	0	1	1.000

S5. Ensemble-member and shell-flag sensitivity

The near-shell MAE contrast is the subgroup mean error among flagged nuclei minus that among all other nuclei. A separate 95% chain-bootstrap interval resamples the same whole-Z blocks, recomputes both means within each draw and preserves the original shell flags. All 10,000 draws contained both subgroups for the four cohorts. The interval does not establish that shell proximity causes the difference.

Table 6: Near-minus-other shell-flag MAE contrast (MeV) and 95% Z-chain percentile interval.

Cohort	n_{near}	MAE contrast	block 95% CI
Development, primary	12	0.600	[0.045,1.182]
Development, broad	21	0.422	[0.044,0.820]
Confirmation, primary	9	0.420	[0.274,0.569]
Confirmation, broad	19	0.413	[0.213,0.631]

Table 7: Leave one mass model out and recompute the remaining three-model mean, sample spread and score.

Cohort	Excluded	n	MAE	ρ	p	Cov. D	Cov. $2D$
Development, primary	FRDM2012	32	0.366	0.364	0.040	0.656	0.969

Cohort	Excluded	n	MAE	ρ	p	Cov. D	Cov. $2D$
Development, primary	HFB27	32	0.449	0.303	0.092	0.562	0.906
Development, primary	DZ31	32	0.372	0.282	0.119	0.656	0.938
Development, primary	WS4	32	0.475	0.364	0.041	0.719	0.844
Development, broad	FRDM2012	58	0.454	0.282	0.032	0.603	0.862
Development, broad	HFB27	58	0.472	0.435	0.001	0.638	0.897
Development, broad	DZ31	58	0.417	0.374	0.004	0.690	0.914
Development, broad	WS4	58	0.519	0.250	0.059	0.672	0.828
Confirmation, primary	FRDM2012	51	0.249	0.320	0.022	0.745	0.941
Confirmation, primary	HFB27	51	0.261	0.265	0.060	0.725	0.941
Confirmation, primary	DZ31	51	0.281	0.332	0.017	0.725	0.980
Confirmation, primary	WS4	51	0.287	0.164	0.250	0.725	0.922
Confirmation, broad	FRDM2012	71	0.299	0.346	0.003	0.718	0.930
Confirmation, broad	HFB27	71	0.321	0.294	0.013	0.690	0.930
Confirmation, broad	DZ31	71	0.331	0.350	0.003	0.690	0.958
Confirmation, broad	WS4	71	0.357	0.246	0.039	0.704	0.901

Table 8: Leave one near-magic case out of the descriptive $E = \beta_0 + \beta_D D + \beta_S I_{\mathrm{shell}}$ regression.

Removed	n	n_{near}	β_0	β_D	β_S	$\mathrm{SE}_{\mathrm{HC3}}$
none	51	9	0.111	0.150	0.404	0.086
44Cr	50	8	0.127	0.110	0.382	0.089
46Mn	50	8	0.106	0.160	0.411	0.094
48Fe	50	8	0.101	0.173	0.415	0.092
70Co	50	8	0.115	0.138	0.417	0.100
52Ni	50	8	0.110	0.150	0.402	0.098
77Cu	50	8	0.106	0.160	0.397	0.103
132Cd	50	8	0.114	0.142	0.356	0.074
101In	50	8	0.110	0.150	0.405	0.096
220Np	50	8	0.149	0.058	0.464	0.070

S6. Nuclear-chart conditions

The frozen distance-tercile bounds are $1.0, 1.0000000000000002, 1.0000000000000004, 2.0$. Because the nearest-neighbor d_2 distribution contains a large tie at exactly 1, the nominal middle category is empty in development and confirmation. The "near" and "far" labels must not be interpreted as equal-sized terciles. Continuous associations avoid this degeneracy.

Table 9: Spearman associations of chart coordinates with absolute error E and spread D.

Cohort	Covariate	Outcome	n	ρ	p
Development, primary	d2	E	32	0.297	0.099
Development, primary	d2	D	32	0.310	0.085
Development, primary	d1	E	32	0.294	0.103
Development, primary	d1	D	32	0.308	0.087
Development, primary	isospin	E	32	0.452	0.009
Development, primary	isospin	D	32	0.323	0.071
Development, broad	d2	E	58	0.283	0.031
Development, broad	d2	D	58	0.432	0.001
Development, broad	d1	E	58	0.278	0.035
Development, broad	d1	D	58	0.419	0.001
Development, broad	isospin	E	58	0.442	0.001
Development, broad	isospin	D	58	0.490	< 0.001
Confirmation, primary	d2	E	51	0.048	0.737
Confirmation, primary	d2	D	51	0.007	0.959
Confirmation, primary	d1	E	51	0.048	0.738
Confirmation, primary	d1	D	51	0.005	0.970
Confirmation, primary	isospin	E	51	-0.030	0.836
Confirmation, primary	isospin	D	51	-0.015	0.916
Confirmation, broad	d2	E	71	0.170	0.157
Confirmation, broad	d2	D	71	0.149	0.215
Confirmation, broad	d1	E	71	0.157	0.192
Confirmation, broad	d1	D	71	0.139	0.247
Confirmation, broad	isospin	E	71	0.140	0.244
Confirmation, broad	isospin	D	71	0.167	0.165

Table 10: Frozen nominal distance-tercile counts and mean error, with the empty middle category made explicit.

Cohort	Label	n	MAE (MeV)
Development, primary	near	30	0.359
Development, primary	middle	0	—
Development, primary	far	2	1.055
Development, broad	near	46	0.383
Development, broad	middle	0	—
Development, broad	far	12	0.698
Confirmation, primary	near	37	0.241

Cohort	Label	n	MAE (MeV)
Confirmation, primary	middle	0	—
Confirmation, primary	far	14	0.263
Confirmation, broad	near	47	0.277
Confirmation, broad	middle	0	—
Confirmation, broad	far	24	0.353

Development-fixed neutron-excess terciles. Table 11 makes explicit the primary-cohort summaries already provided in the original `nuclear_mass_repo/tables/Table4_Conditional_False_Consensus.csv`. For $I = (N - Z)/A$, the two internal cut points are 0.13954728806213954 and 0.20608627060239953, as recorded in the frozen development configuration. The lower bin includes the first cut point, the middle bin includes the second, and the exterior bounds are open. No cut point is recomputed from confirmation. These descriptive summaries complement the continuous isospin associations above and do not alter the headline analysis.

Table 11: Primary-cohort neutron-excess terciles using the unchanged development-defined cut points.

Cohort	I bin	n	MAE (MeV)
Development, primary	Low	11	0.160
Development, primary	Middle	10	0.160
Development, primary	High	11	0.866
Confirmation, primary	Low	10	0.423
Confirmation, primary	Middle	13	0.122
Confirmation, primary	High	28	0.242

S7. Calcium two-neutron separation energies

The $Z = 20$, $N = 28$–40 comparison reports the fully reproducible experimental AME2020, four global-model predictions and seven published S_{2n} values from Coraggio *et al.*, Phys. Rev. C **102**, 054326 (2020), Table II. The tabulated experimental uncertainty is the quadrature propagation of the two tabulated mass-excess errors, *not* the full evaluated covariance; the latter was not provided in the processed release. The five eligible AME2020 points $A = 48$–56 give an absolute-error mean of 0.719 MeV for the published Coraggio calculation versus 1.375, 0.909, 0.896 and 0.333 MeV for the four global models. This is a small retrospective illustration and does not establish relative predictive skill. Error scores use the *recomputed AME2020* reference throughout, not the earlier experimental values reproduced in the 2020 paper. The GSM reconstruction, documented amendment and common-isotope comparison are described below.

Table 12: Two-neutron separation energies S_{2n} (MeV) in the calcium chain.

Nuclide	N	AME2020	approx. σ	FRDM2012	HFB-27	DZ31	WS4	Coraggio '20
48Ca	28	17.228	0.002	14.993	15.863	16.403	16.444	18.105
50Ca	30	11.507	0.002	10.963	11.723	11.123	12.001	12.054
52Ca	32	10.820	0.002	8.613	9.803	8.693	10.552	11.754
54Ca	34	7.037	0.048	7.263	7.533	6.993	6.929	6.914
56Ca	36	4.492	0.254	6.153	5.943	5.593	4.482	3.377
58Ca	38	—	—	4.973	5.543	4.463	3.665	3.347
60Ca	40	—	—	4.433	2.213	3.583	3.247	3.017

S7.1. GSM vector reconstruction and amendment

The RQ3 question and the $A = 48$–60 region were specified in the original plan. The data-source rule was amended during revision to admit vector-digitized values, after the extracted values had been inspected and after requests for original outputs had remained unanswered. This amendment is recorded separately; the frozen protocol is not rewritten. The data are reconstructed plotted estimates, not author-supplied numerical outputs. They are neither added to the four-model global ensemble nor used to alter the primary mass cohorts or thresholds.

The journal PDF (page 5) and arXiv:2008.11948v1 (page 11) were checked separately. Five major ordinate ticks ($-5, 0, 5, 10, 15$ MeV) calibrate the linear coordinate transformation. Abscissae are matched to the mass-number ticks, and red polyline vertices are checked against triangle centroids. All 15 values agree after rounding to 0.01 MeV. Maximum journal-versus-arXiv disagreement is 0.00020 MeV; the rounded CSV differs from the journal coordinates by at most 0.00461 MeV. The maximum journal axis-fit residual is 0.00510 MeV, and marker-centroid versus line-center disagreement is 0.00481 MeV. These diagnostics check graphical reconstruction; they do not measure the unknown precision of the original model outputs.

Table 13: GSM S_{2n} reconstructed from Li et al., Phys. Rev. C 102, 034302 (2020), Figure 2(b), in MeV. Rows $A = 62$–72 are an archived extension outside the planned RQ3 range. A dash means no value in the source figure; ^{48}Ca is not imputed.

A	N	^{48}Ca core	^{54}Ca core
50	30	11.68	—
52	32	12.14	—
54	34	8.02	—
56	36	3.95	4.37
58	38	3.09	3.01
60	40	2.24	1.86
62	42	—	0.48
64	44	—	0.54
66	46	—	0.78
68	48	—	0.97
70	50	—	1.29
72	52	—	-2.28

S7.2. Comparison on the same measured isotopes

All six rows below use exactly $A = 50, 52, 54, 56$ and the same recomputed AME2020 references. This avoids comparing the four-point GSM series with five-point statistics that include ^{48}Ca. The GSM ^{54}Ca-core calculation has only one measured overlap (^{56}Ca), with absolute residual 0.122 MeV; it is kept separate. No measured-reference error is assigned to ^{58,60}Ca, whose AME2020 masses are estimated.

Table 14: Descriptive errors on ^{50,52,54,56}Ca ($n = 4$). All entries are in MeV. Bias is calculation minus AME2020. GSM scores use the two-decimal figure-derived values.

Model	MAE	RMSE	Bias
FRDM2012	1.159	1.412	-0.216
HFB-27	0.795	0.926	+0.286
DZ31	0.914	1.213	-0.364
WS4	0.220	0.286	+0.027
Coraggio (Table II)	0.680	0.780	+0.061
GSM (^{48}Ca core)	0.755	0.871	+0.483

S7.3. Sensitivity to the extraction allowance

For figure estimate g_i and fixed experimental reference e_i, we vary each g_i independently within $g_i \pm \delta$. The smallest and largest possible MAEs under this stress test are

$$\frac{1}{n}\sum_i \max(|g_i - e_i| - \delta, 0), \qquad \frac{1}{n}\sum_i (|g_i - e_i| + \delta).$$

The code also supplies the analogous RMS limits and results for $\delta = 0, 0.005, 0.01, 0.05, 0.10$ MeV. For the four-point ^{48}Ca-core series, MAE is 0.755 MeV and its range at $\delta = 0.10$ MeV is 0.655–0.855 MeV. Coraggio's 0.680-MeV MAE lies within that range, so their small nominal difference is not robust to the adopted allowance. The ^{54}Ca-core ^{56}Ca residual has range 0.022–0.222 MeV. These are conditional sensitivity ranges, not confidence intervals, theory-error bars or guaranteed bounds on the original solver values. Experimental uncertainty remains a separate quantity.

The ^{54}Ca-core minus ^{48}Ca-core differences at $A = 56, 58, 60$ are $+0.42, -0.08, -0.38$ MeV. Neither an average of the two core calculations nor a selection based on agreement with data is used. Both regional studies are retrospective, and Li et al. used experimental ^{55}Ca information in setting their potential. This case study therefore does not provide an independent calibration or a causal demonstration of the origin of global-model errors.

S8. Implementation and limitations

The independently written analysis is implemented in Python with NumPy, pandas, SciPy and Matplotlib. No original scoring routine is imported, and the primary 51 mass and 34 separation-energy rows are asserted numerically against the archived scores. The three-model leave-one-out calculations, quadrature combination with experimental uncertainties, percentile and chain bootstrap for MAE and coverages, CRPS, shell regression and calcium global-model values are reproducible from the supplied tables. Published calcium-shell-model values are independently checked against Coraggio *et al.*'s Table II using the provided validation script; the source-paper PDF is not redistributed. The GSM coordinate audit can be reproduced offline; a fresh source-PDF check requires separately obtained PDFs and PyMuPDF. The DZ31 source distribution, Windows reproduction bundle, original run report, raw output and independent verification scripts remain included. The working Gaussian CRPS exercise is *not* an ensemble posterior predictive probability: its scale is estimated from the development data, and its transfer is checked on confirmation.

This revision package postdates the stated frozen protocol. Without externally verifiable pre-registration, we refer only to a *chronology-respecting analysis with a prespecified plan*. Experiment dates inferred from AME membership do not establish that each model was fitted before every added mass. The sample is clustered, short-range and not representative of remote neutron-rich nuclei.

Files for exact-value replication:

- `cohort_metrics.csv`
- `cohort_Zblock_coverage_MAE_v9.csv`
- `quintiles_with_bootstrap.csv`
- `confirmation_71_nuclei.csv`
- `leave_one_model_out.csv`
- `near_magic_leave_one_out.csv`
- `chart_condition_associations.csv`
- `calcium_global_model_S2n.csv`
- `calcium_GSM_matched_metrics.csv` and `calcium_GSM_matched_residuals.csv`
- `calcium_GSM_extraction_sensitivity.csv` and `calcium_GSM_core_comparison.csv`
- `raw_source_audit/gsm_vector/`: CSV, coordinates, verification records and source checker
- `GSM_PROTOCOL_AMENDMENT.txt`
- `raw_source_audit/DZ31_verification_report.txt`
- `raw_source_audit/DZ31_recovered_raw_verification_v9.json`
- `raw_source_audit/DZ31_windows_run_provenance.md`
- `raw_source_audit/DZ31_Windows_Verification.zip`
- `raw_source_audit/DZ31_windows_raw_output/massex.out`
- `raw_source_audit/verify_recovered_massex_v9.py`